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1. Executive Summary

SomaCampus is a school operating system for teachers and principals, built on React 19, Vite, Tailwind CSS and Supabase, and piloted at Grace's Cambridge Centre in Kampala. This review combines a full codebase audit with black-box GUI testing of the live application in a real browser, executed as the four everyday end users: teacher, principal, parent and student, using the seeded demo accounts.

The verdict is two-sided. The product looks and feels premium: the visual design system, role-aware navigation, and the daily-attendance workflow are genuinely better than most school software. The engineering culture is equally strong - the repository's own verification gate passes end to end, with database-contract audits, migration audits, TypeScript checks and 840 automated tests all green. But the product is not yet ready for its users: two of the six role homepages are broken in the live demo, the teacher's most important call-to-action dead-ends in an error, the principal's timetable publish flow is blocked by a missing database policy, and several dashboards present self-contradicting numbers that would erode a principal's trust within the first week.

840 / 840 automated tests passing (verify:all gate)	14 screens and flows GUI-tested in a live browser	4 P0 broken journeys found in live testing
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All four blocking failures are integration- and data-contract issues rather than architectural ones, which is why they survived the automated gate: the unit and contract suites validate components and schema assumptions in isolation, while the broken journeys only manifest when a real Supabase session meets the seeded data. The recommended fix order in Section 8 is calibrated accordingly: restore the four dead ends, make the numbers agree, then run a product information-architecture pass for teachers.

2. Scope and Method

2.1 Codebase review

The audit covered the application entry and routing (src/App.tsx), the authentication and role-resolution layer (src/lib/authContext.tsx, server-side role RPC with fail-closed behaviour), the design system (src/components/ui and layout), all 25 modules under src/modules, 71 Supabase migrations, and the verification tooling in .agents. The product spans six roles - teacher, principal, admin, bursar, parent and student - with modules for attendance, lesson delivery, assignments, timetable building with a constraint solver, HR and payroll, fees, expenses, inventory, communication and parent or student portals.

2.2 GUI test protocol

Black-box testing was executed in a real browser at 1440 by 900 against the development server and the live Supabase backend. Every conclusion is based on observed page behaviour, cross-validated between the accessibility tree and screenshots. Fourteen test points were executed across four personas; every screenshot referenced in this report is preserved in the gui-test-screenshots directory of the repository.

Environment preparation disclosure: the Vite dev server was started (port 5173) and seeded demo accounts were used exclusively. One write was performed during testing - marking one student absent in the Stage 5 Blue register - which is the flow under test and is audit-recorded by design. No code was modified during testing.

2.3 Demo credentials used

Persona	Account	Landing page
Teacher - Sarah Namukasa	teacher@somacampus.ug	/teacher/today
Principal - Dr. Edward Ssenyonga	principal@somacampus.ug	/dashboard/school
Parent - Florence Kyomugisha	parent@somacampus.ug	/parent/home
Student - Amari Kyomugisha	student@somacampus.ug	/student/home

Table 1 - Seeded demo accounts exercised during the test pass.

3. What Works Well

3.1 Login and persona entry

The login screen sets the tone: clear brand identity, a descriptive card per role, and one-click sign-in for each verified demo persona. A teacher or principal needs no training to get in. This is a strong first impression and worth protecting.

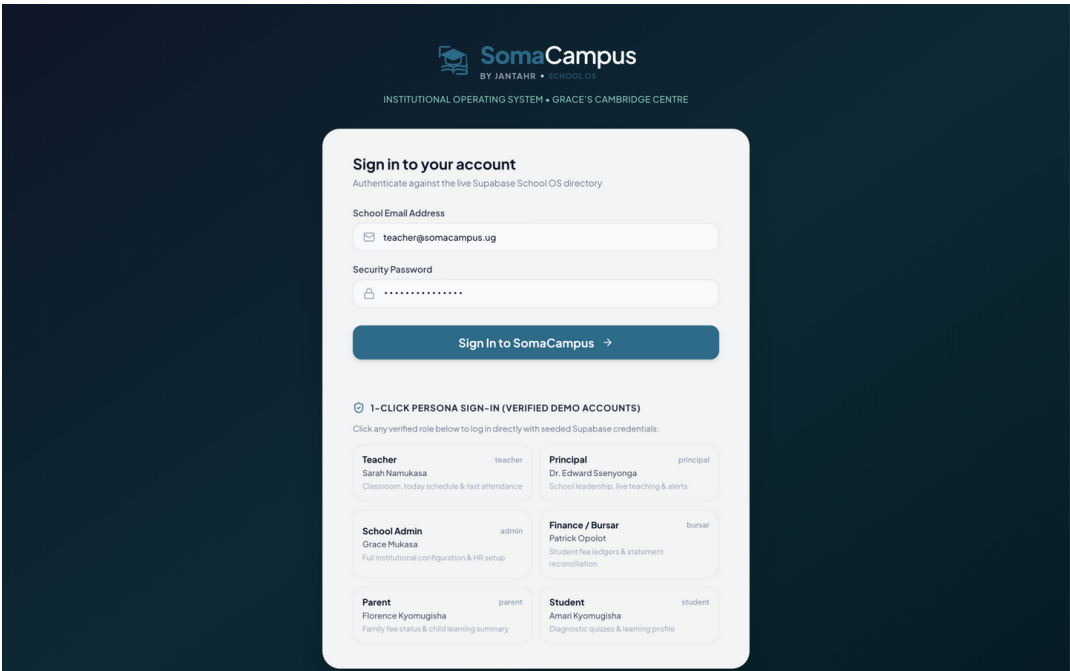


Figure 1 - Login screen with one-click verified demo personas.

3.2 Daily attendance register (the gold standard flow)

The class register is the best flow in the product. Tapping Present, Absent, Late or Excused updates live counters; editing an already-recorded day reveals an "Attendance Audit Mode" notice and an optional reason field; the save button shows an honest disabled spinner; and the dashboard card refreshes to the new tally (verified 9 present, 1 absent after save). Every register in the product should match this standard.

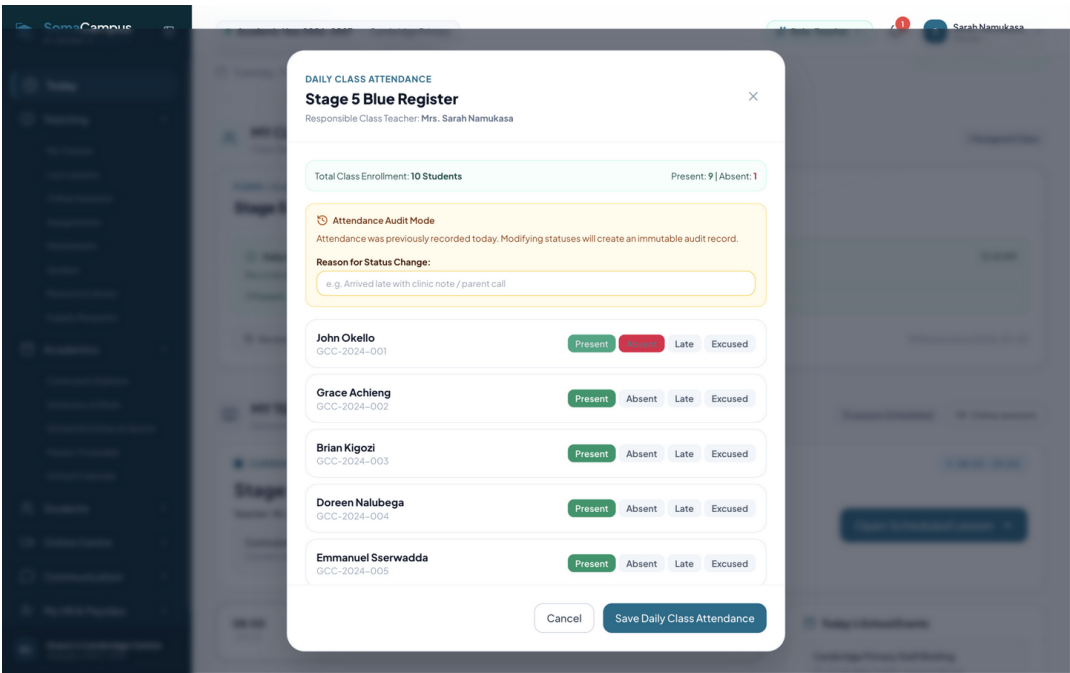


Figure 2 - Register with audit mode, reason field and live counters.

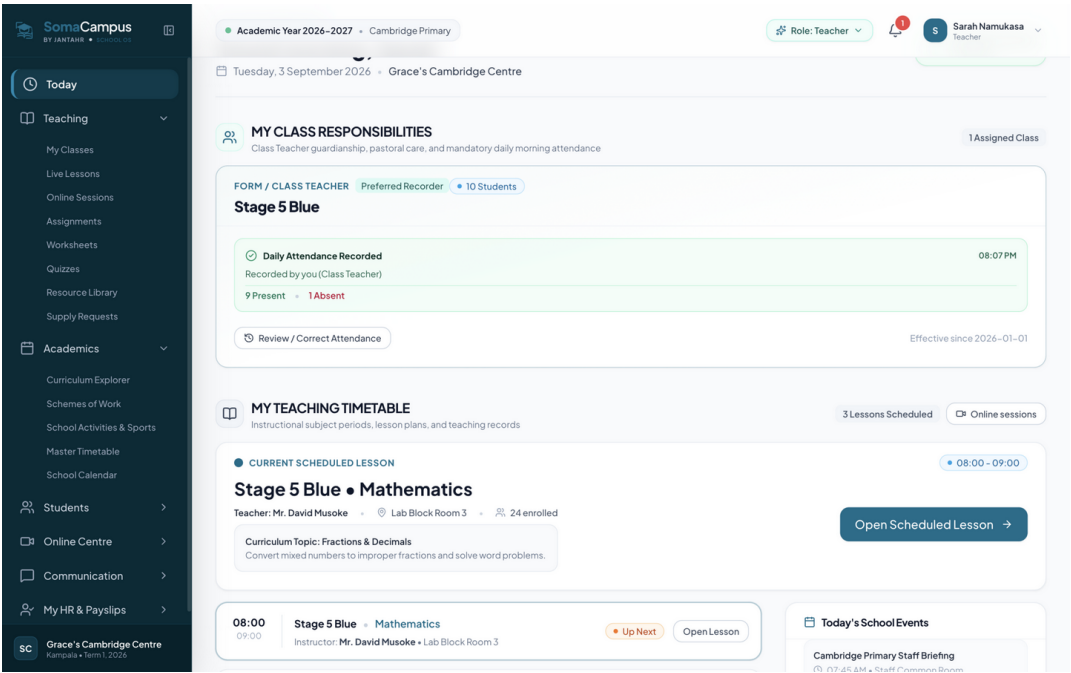


Figure 3 - Dashboard card updated after saving: 9 present, 1 absent.

3.3 Principal monitoring surfaces

The School Cockpit, Live Lessons Monitor and Fee Accounts render cleanly and load fast. The Live Lessons Monitor in particular does its one job well: expected versus submitted periods for the day, filters, and an expandable note per period. The data-accuracy issues affecting these screens are covered in Section 5; the interaction design itself is sound.

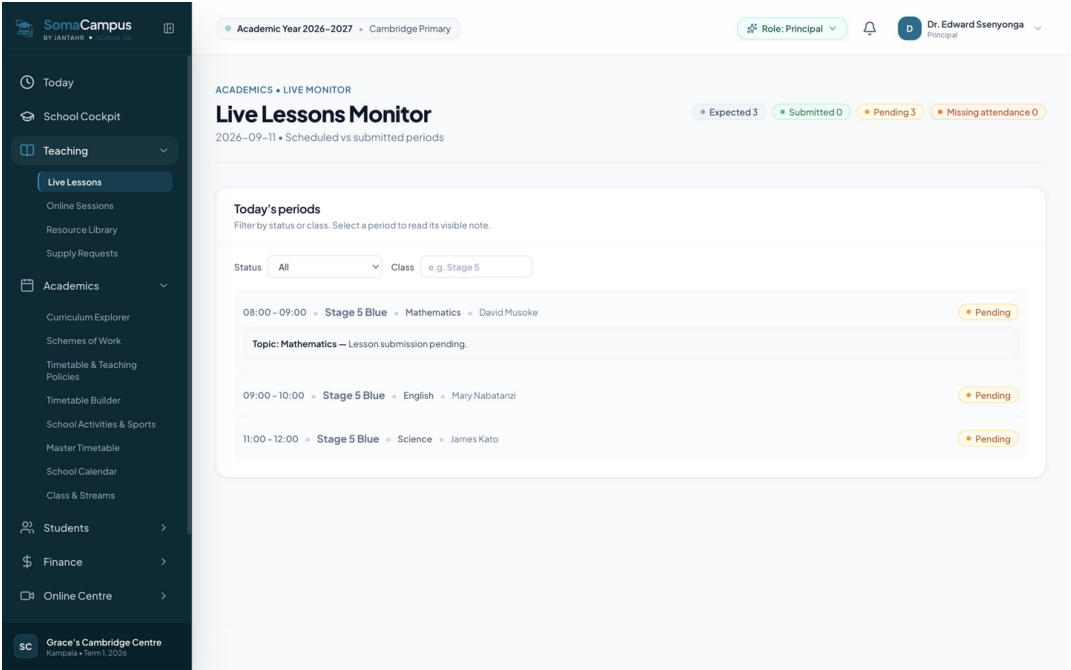


Figure 4 - Live Lessons Monitor with an expanded period note.

4. Critical Failures (P0)

Four journeys are broken end-to-end for ordinary users. Each is reproduced below with evidence and a code-level starting point.

4.1 F1 - Lesson Cockpit fails from the teacher's primary call-to-action

From the Teacher Today page, clicking "Open Scheduled Lesson" - the single most important teacher journey - lands on a URL of the form `/teaching/classes/5555.../lessons/tt-entry-001` and renders "Lesson cockpit unavailable: Could not load lesson context for timetable entry tt-entry-001." The classroom cockpit is where lesson delivery, attendance and evidence capture live, so this dead-end blocks the core teaching loop.

Starting point: the identifier reaches `src/modules/teaching/lessonService.ts` (line 101, which raises "Could not load lesson context for timetable entry") but its context query fails against live data. The fixture-era identifier `tt-entry-001` originates in `src/modules/teacher/fixtures/teacherFixtures.ts`; the live path should be verified to pass real timetable entry identifiers and to satisfy the cockpit's joins.

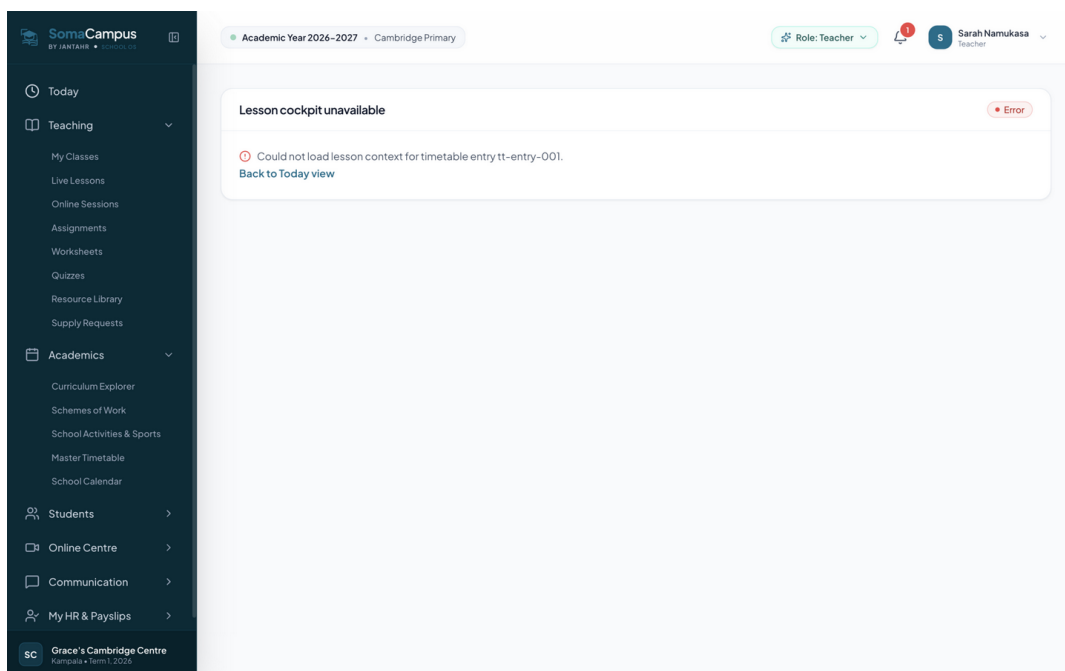


Figure 5 - Lesson Cockpit error after clicking the main CTA.

4.2 F2 - Parent portal cannot load the child overview

The Family Portal home (/parent/home) shows both a red error banner and an empty state at the same time - redundant double-failure messaging - and the failure persists across retries. For a parent, this screen is the whole product. Note that the page's fail-closed behaviour is correct and deliberate (src/modules/parent/ParentHomePage.tsx line 207); the defect is that the seeded demo data never satisfies the guardian resolution chain, so the flagship parent journey was never demonstrably green.

Starting point: the resolver walks auth user, then people.auth_user_id, then student_guardians, then active student_enrolments (src/modules/auth/parentIdentity.ts). Verify that the parent@somacampus.ug auth user is linked to a person row with a student_guardians row for Amari Kyomugisha, and that RLS grants the select.

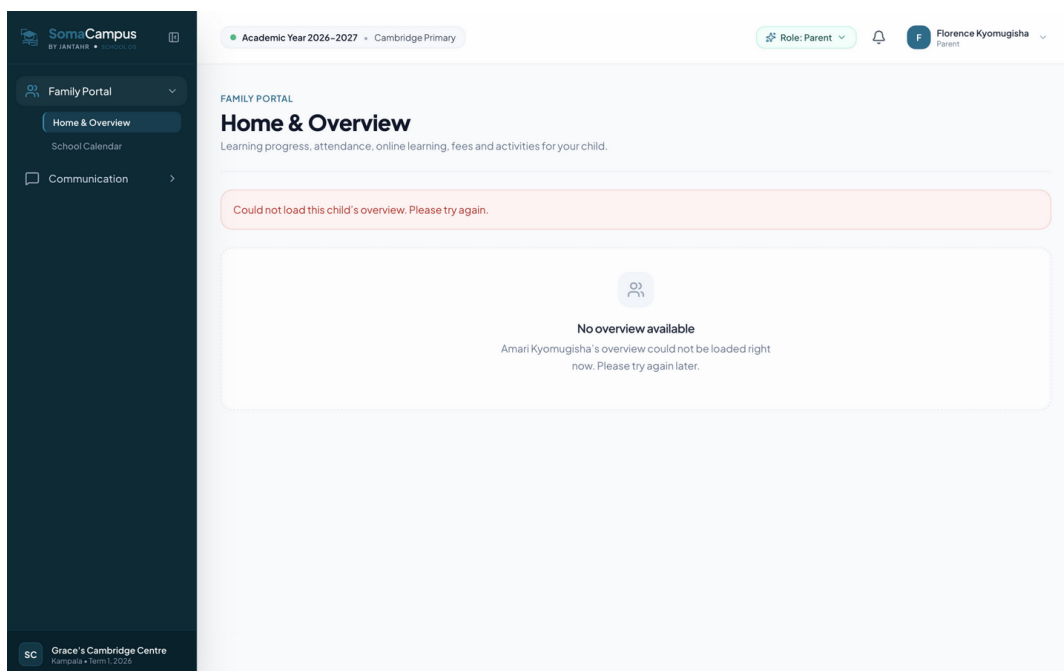


Figure 6 - Parent portal: error banner plus empty state shown together.

4.3 F3 - Student portal leaks a raw database error to a child

The student home (/student/home) renders the literal database engine error "JSON object requested, multiple (or no) rows returned" in a red banner. Showing a PostgreSQL PGRST116 message to a primary-school student is the exact class of leak the repository's own Production Trust Gate forbids.

Starting point: src/modules/online/StudentOnlineHomePage.tsx (line 71) assigns err.message directly to the visible load error. The service call beneath it performs a single-row lookup that returns zero or multiple rows for the demo student. Map engine errors to friendly copy, log the detail to the console, and fix the underlying lookup.

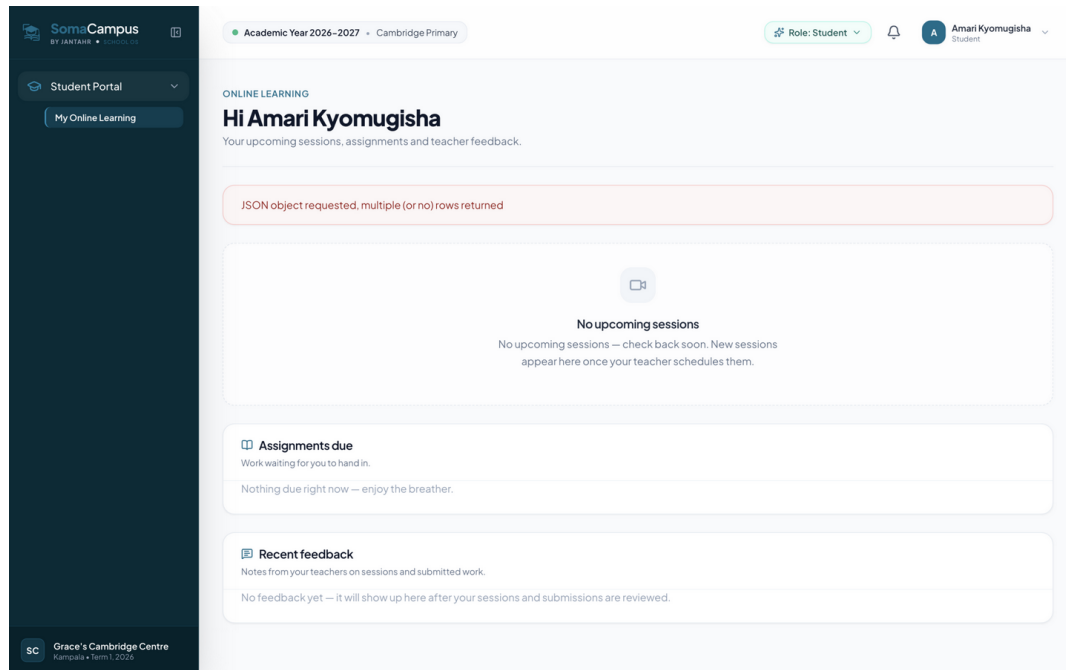


Figure 7 - Student home showing the raw PostgREST error string.

4.4 F4 - Timetable publishing is blocked by a missing database policy

The principal's Timetable Wizard completes all four guided steps - foundation check, staffing auto-fill, schedule generation, and week-grid review - but clicking "Publish timetable" fails with "new row violates row-level security policy for table timetables." The entire planning journey therefore dead-ends at its final step. This finding was independently reproduced by a second end-to-end test pass of the wizard and verified against the migration history during this review.

Root cause, verified: the timetables and timetable_entries tables have exactly one policy each - timetables_auth_read and timetable_entries_auth_read, both FOR SELECT only (supabase/migrations/20260903000000_somacampus_core_schema.sql lines 368-369). No later migration adds INSERT or UPDATE policies for either table. The wizard's publish handler (src/modules/planning/TimetableWizard.tsx, publishAll, line 272) performs a direct insert into timetables and then into timetable_entries, which PostgreSQL fail-closes.

A corroborating signal from this review's own test session: the Master Timetable page displays "Status: PUBLISHED" with 15 periods scheduled, yet the UI publish path is impossible - so the published state can only have arrived through seed data or service-role writes, never through the product itself.

Recommended fix - use the atomic RPC, not broader policies. The codebase already contains approve_timetable_atomic and publish_timetable_atomic as SECURITY DEFINER functions with in-function leadership checks (migration 20260916000000). The wizard bypasses its own architecture by inserting directly and only then calling the RPCs. Moving timetable and entry creation into an atomic RPC keeps RLS fail-closed, closes the orphaned-timetable window between the two direct inserts, and avoids maintaining leadership gating in two places. Adding direct INSERT/UPDATE policies would also unblock the button but is the weaker option.

5. Data-Trust Defects (P1)

These screens render, but the numbers contradict each other. For school staff, a number they cannot trust is worse than no number at all. Each item below was observed in a single session on the live demo data.

#	Where	Defect observed	Anchor
D1	Principal cockpit KPI row vs its own chart	Student Attendance reads 0% with a "-90.0% this week" delta while the adjacent chart shows bars near 90-100; Active Teaching Staff reads "6, 0% staff present". A negative percentage delta is also nonsensical formatting.	SchoolDashboard Page
D2	Teacher Today date	Header shows "Tuesday, 3 September 2026" on Friday 11 September 2026. The label is hardcoded in the live code path, together with hardcoded school events.	teacherService.ts: 370
D3	Teacher Today timetable	"MY TEACHING TIMETABLE" lists lessons taught by other teachers (David Musoke, James Kato) and says "24 enrolled" while the class card on the same page says 10 students.	teacherService.ts / fixtures
D4	Attendance card vs register modal	Before the test edit, the class card reported 7 present / 1 absent while the register modal reported 10 present / 0 absent - two different truths one click apart.	attendance service
D5	Fee Accounts ledger	A student with UGX 0 assessed, UGX 0 paid and UGX 0 balance is flagged "Overdue". No-data is being treated as arrears.	fees clearance logic
D6	Assignments list	Records named TEST_Audit_Score_Assignment, TEST_Principal_Override_Assignment and TEST_Math_Fractions_HW are visible to teachers in the live list.	demo data hygiene
D7	Time formats	Three formats on one page: "12:43 AM" (fixture stamp), "08:07 PM" (real save) and 24-hour "07:55" (clock-in).	display layer

Table 2 - Observed data-trust defects with code or data anchors.

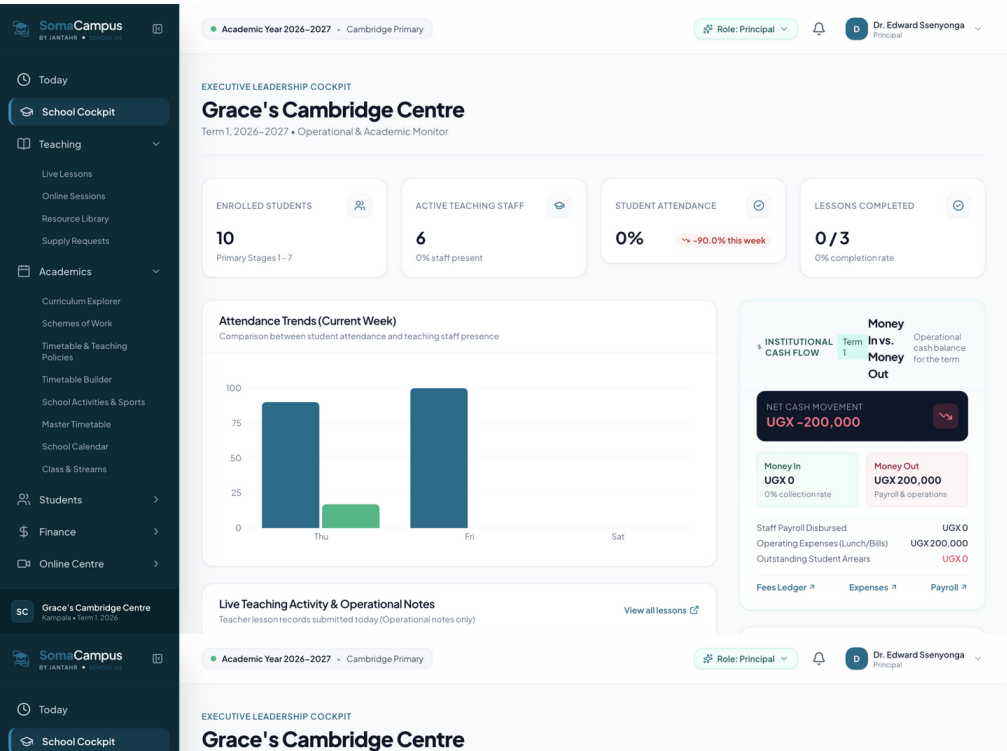


Figure 8 - Principal cockpit: KPI row contradicts the chart below it.

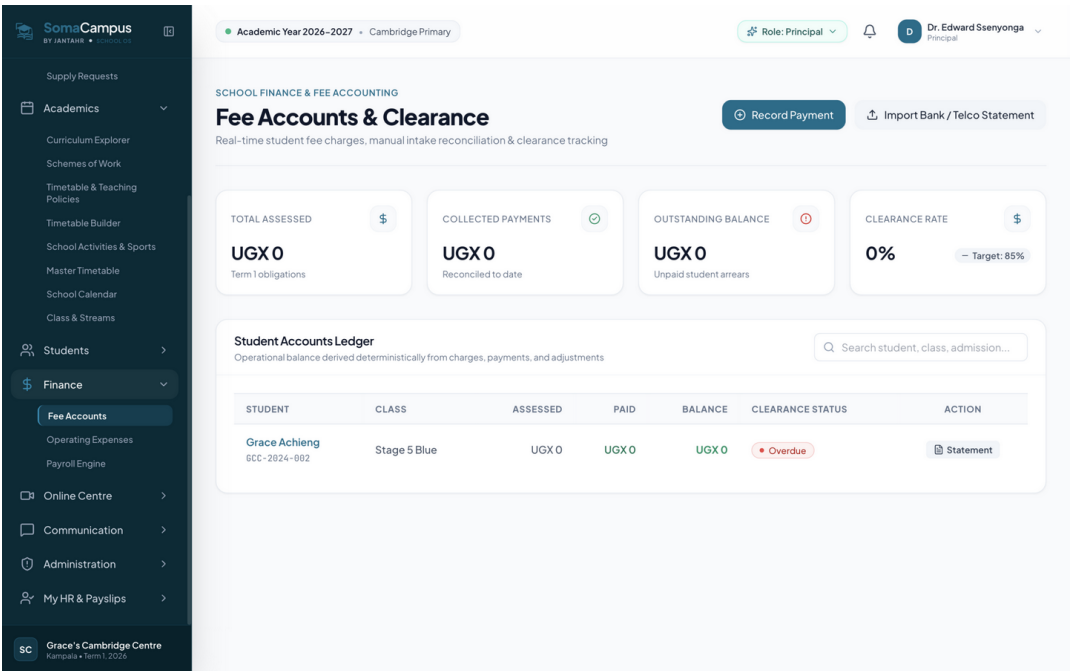


Figure 9 - Fee ledger: zero-balance account flagged Overdue.

6. Product and UX Review (Teacher and Principal Lens)

6.1 Strengths worth protecting

- Visual design is top-notch. The dark-teal navigation rail, light workspace, Plus Jakarta Sans typography and a consistent card and status-pill system read premium in every screenshot; nothing looked misaligned or cheap.
- Information architecture matches the school day. Teachers land on Today with clock-in, class responsibilities, the current lesson and events; principals get a cockpit, live lessons and finance. Role-based navigation filtering works - teachers never see payroll.
- The attendance register is the UX gold standard for the whole product (Section 3.2), and the four-step Timetable Wizard with auto-fill, week grid, bell bands and in-grid slot editing is a major step forward for principals - once the publish blocker (F4) is fixed.
- An honest engineering culture is real: fail-closed services, server-side privacy gating, archive-not-delete, honest placeholders for unbuilt features, and an 840-test gate.

6.2 Concerns for the target users

- Teachers are shown engineering, not their job. The teacher's Master Timetable page opens with "Teaching Allocation Architecture ... Deterministic CSP Solver", allocation approval steps and a "Run Constraint Solver" button. Allocation and solving are principal or admin work; a teacher needs "my week, my classes, my room changes". A read-only personal week grid would serve teachers far better. The same double-header problem affects principals in the wizard itself, where the legacy three-step summary cards and the wizard's own breadcrumb stack two navigation systems on one screen - a finding made independently by both test passes in this review.
- Navigation breadth: roughly 18 destinations for a teacher, with Worksheets and Quizzes as placeholders in the primary group. Consider demoting unbuilt items.
- Dev scaffolding leaks into user space: the green "Role: teacher" chip in the header is a development persona switcher, and the "Academic Year 2026-2027 - Cambridge Primary" context chip is hardcoded and will rot.
- Accessibility gaps: the 40 register status toggles are unnamed buttons in the accessibility tree (statuses are not associated with student names), the register close button has no accessible name, and the selected "Absent" chip is dark red on dark red with poor contrast.
- Login cards promise more than the product delivers: "Diagnostic quizzes and learning profile" leads to an empty shell for the student persona. Set expectations or build a minimal real version.

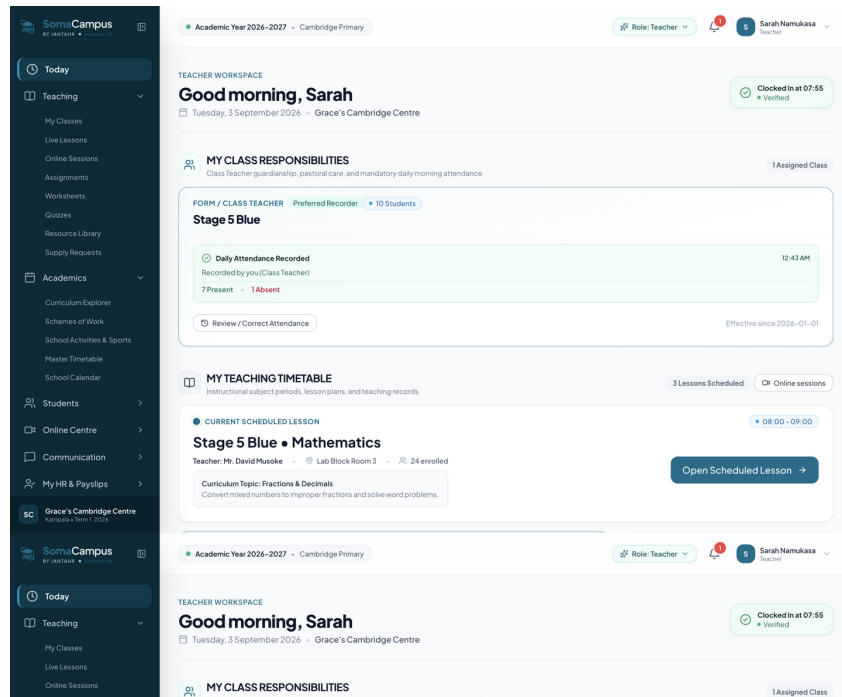


Figure 10 - Teacher Today: strong layout, but stale date and mixed instructor names on "MY TEACHING TIMETABLE".

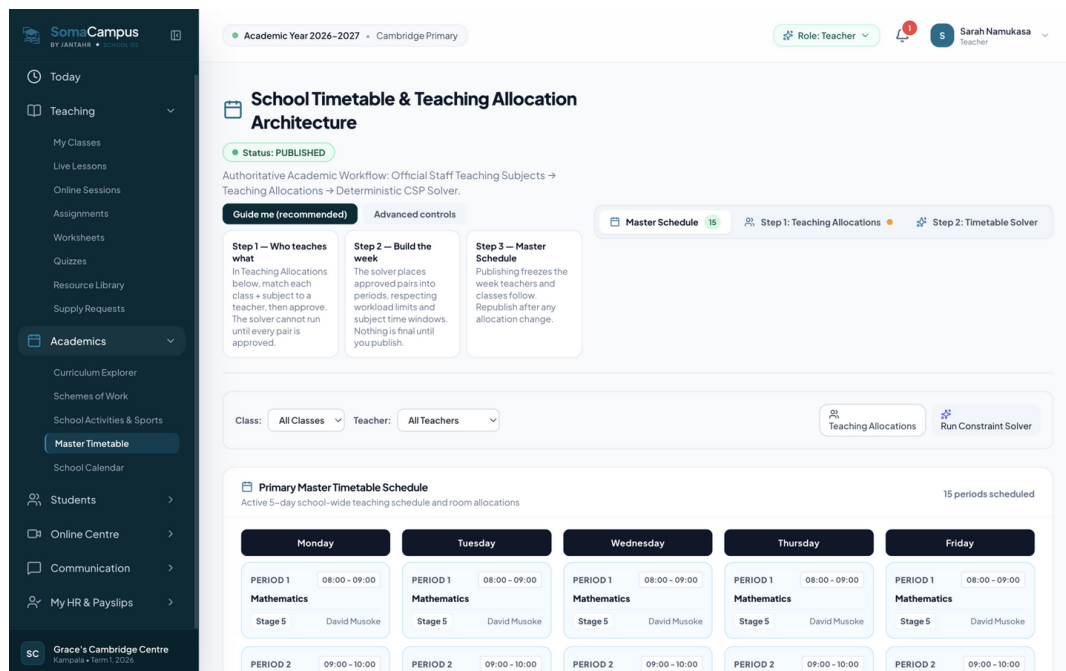


Figure 11 - Master Timetable: solver jargon presented to teachers.

7. Code Health

7.1 Verification gate

npm run verify:all - PASS. Database-contract audit, migration audit, TypeScript checks, and the Vitest suite: 94 test files, 840 tests passed, 26 skipped (live-database tests). Mock-rot, contract and migration scanners all report clean.

7.2 Findings

- Fixture-live boundary is blurred: INITIAL_TEACHER_SCHEDULE in src/modules/teacher/fixtures/teacherFixtures.ts backs the mock path, and fixture-era identifiers such as tt-entry-001 exist in the live database - the direct source of F1.
- Error surfacing pattern: several pages assign err.message directly to visible error banners; the student portal proves the risk. Standardise on friendly copy plus logged detail.
- Oversized files (maintainability): InventoryPage.tsx (2,113 lines), types/domain.ts (1,685), TimetableDraftPage.tsx (1,616), StudentDetailPage.tsx (1,473), and nine further files above 800 lines.
- No route-level code splitting: the entire app is one eager bundle (about 81k lines of TypeScript), which teachers on tablets or constrained networks will feel on first load.
- Persona names in the development switcher (for example "Dr. Florence Namugga") do not match the seeded users ("Dr. Edward Ssenyonga") - a small but confusing demo detail.
- Zero TODO or FIXME markers - unusually clean; incompleteness is tracked honestly through placeholder modules and phase documents instead.

8. Recommended Fix Order

Priority	Work item	Why first
P0	Restore the four dead ends: Lesson Cockpit context for live timetable entries; parent-to-child seed linkage (student_guardians plus auth_user_id for the demo parent); map PGRST116 on the student home to friendly copy; and route timetable publish through an atomic create-and-fill RPC so RLS stays fail-closed (Section 4.4).	These are the journeys teachers, parents and principals take first.
P1	Make the numbers agree: cockpit attendance and staff KPIs versus chart, enrolment counts (10 versus 24), attendance card versus register, fees "Overdue" on zero accounts, real date in teacherService.ts line 370.	One contradiction costs more trust than one missing feature.
P1	Purge TEST_* rows from the demo database and add a seed-validation rule against the TEST_ prefix.	Test pollution in a teacher's live list reads as unprofessional.

Priority	Work item	Why first
P2	Product IA pass for teachers and principals: collapse the legacy solver header when the guided wizard is active (confirmed independently by two test passes), give teachers a personal week view, move solver and allocation controls behind principal or admin, demote placeholder nav items.	Sharpens the product for its primary daily users.
P2	Accessibility sweep (button naming, chip contrast) and one timestamp format everywhere.	Removes friction for assistive-tech users and standardises reading.

Table 3 - Remediation roadmap in recommended order.

The dev server used for testing was left running at <http://localhost:5173>, and all fourteen screenshots are preserved under `gui-test-screenshots` in the repository. No production code was changed during this review.

Appendix A. GUI Test Point Matrix

#	Test point	Result	Evidence
T0	Login page and one-click personas	PASS	t0_login.png
T1	Teacher Today dashboard loads	PASS (data issues, see D2/D3)	t1_teacher_today.png
T2	Lesson Cockpit from "Open Scheduled Lesson"	FAIL	t2_lesson_cockpit_error.png
T3	Attendance register modal opens	PASS (mismatch vs card, D4)	t3_attendance_register.png
T4	Mark student absent, audit mode appears	PASS	t4_absent_marked.png
T5	Save attendance with reason	PASS	t5_attendance_saved.png
T6	Dashboard card refreshes after save	PASS	t6_after_save.png
T7	Assignments list and filter tabs	PASS (TEST_rows, D6)	t7_assignments.png
T8	Master Timetable week grid renders	PASS	t8_master_timetable.png
T9	Principal School Cockpit	PASS (KPI contradictions, D1)	t9_principal_cockpit.png
T10	Live Lessons Monitor and period detail	PASS	t10_live_lesson_detail.png
T11	Fee Accounts ledger	PASS (Overdue logic, D5)	t11_fees.png
T12	Parent Home and Overview	FAIL (persistent)	t12_parent_home.png
T13	Student Online Home	FAIL (raw DB error)	t13_student_home.png
T14	Timetable Wizard four-step walkthrough and publish	FAIL at publish (RLS, F4)	code-verified

Table 4 - Fourteen GUI test points across four personas, plus the code-verified wizard publish pass (F4). Screenshots in gui-test-screenshots.

Appendix B. Screenshot Index

File	Moment captured
t0_login.png	Login screen with persona cards
t1_teacher_today.png	Teacher Today dashboard (full page)
t2_lesson_cockpit_error.png	Lesson Cockpit unavailable error
t3_attendance_register.png	Register modal, initial state
t4_absent_marked.png	Absent marked, audit mode and reason field
t5_attendance_saved.png	Save in progress with spinner
t6_after_save.png	Dashboard card updated after save
t7_assignments.png	Assignments list with TEST_ rows
t8_master_timetable.png	Master Timetable and solver controls
t9_principal_cockpit.png	Principal cockpit (full page)
t10_live_lesson_detail.png	Live Lessons Monitor, period expanded
t11_fees.png	Fee Accounts ledger with Overdue flag
t12_parent_home.png	Parent portal failure state
t13_student_home.png	Student portal raw database error

Table 5 - Evidence index. All files are in gui-test-screenshots/ at the repository root.